Fundamental understanding of thermal transport across solid interfaces

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4.1 Introduction

Thermal resistances from interfaces were first observed by Kapitza in 1941 when he studied heat transfer between copper and liquid helium at extremely low temperatures [1,2]. Thermal boundary resistance is then named as Kapitza resistance [2]. At macroscopic length scales, the thermal boundary conductance (TBC, G) is described by the interfacial form of Fourier's law as

$$G = \frac{Q}{\Delta T} \tag{4.1}$$

Here, Q is the heat flux flowing across an interface and ΔT is the temperature difference across the interface. Based on the phonon gas model, a Landauer theory to describe TBC was developed [3–5]:

$$G = \sum\nolimits_p \frac{1}{2} \int\int D_1(\omega) \frac{df_{BE}}{dT} \hbar\omega v_1(\omega) \tau_{12}(\theta, \omega) \cos\theta \sin\theta d\theta d\omega, \tag{4.2}$$

where D is the phonon DOS, f_{BE} is the Bose-Einstein distribution function, $\hbar$ is the reduced Planck constant, ω is the phonon angular frequency, v is the phonon group velocity of material 1, τ_{12} is the transmission coefficient from material 1 to the other side of the interface (material 2), θ is the angle of incidence, and the sum is over all incident phonon modes [4]. There are two main assumptions in the Landauer-based description of TBC that need to be re-checked. One is the phonon gas model that may not hold at interfaces due to the missing of symmetry at heterointerfaces and new localized modes may emerge [6,7]. The other is thermal equilibrium near the interface that is especially not true for high mismatched interfaces [8].

To calculate the transmission coefficient, two classic models were developed at the 1950s: acoustic mismatch model (AMM) and diffuse mismatch model (DMM) [3,9]. In terms of AMM, the interface is treated as a smooth plane, and the phonons are

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governed by continuum acoustics. Here, phonons are considered as plane waves. These assumptions are more likely to hold at extremely low temperatures when phonon wavelengths are more likely larger than the interface roughness [3]. The transmission coefficient from AMM is [3]

$$\tau_{12} = \frac{4Z_1Z_2}{(Z_1 + Z_2)^2}, \tag{4.3}$$

Here, $Z_i = \rho_i C_i$ is the acoustic impedance that is the product of the mass density and the phonon group velocity. i is the material that jointly composes the interface [3].

DMM assumes the interface is rough and an incident phonon forgets its original momentum and direction. The probability of the phonon transmitting through the interface depends on the phonon number of modes on each side of the interface. The expression of the transmission function from DMM [10] is

$$\tau_{12}(\omega) = \frac{\sum_p M_2(\omega)}{\sum_p M_1(\omega) + \sum_p M_2(\omega)}, \tag{4.4}$$

where M is the phonon number of modes. According to DMM, the transmission function is independent from the angle of incidence. Therefore, the Landauer formula can be simplified as

$$G = \sum_p \frac{1}{4} \int D_1(\omega) \frac{df_{BE}}{dT} \hbar\omega v_1(\omega) \tau_{12}(\omega) d\omega. \tag{4.5}$$

This formula is widely used to estimate TBC of real interfaces at room temperature in the literature [8,11–14].

The DMM and AMM were originally developed to explain the thermal resistances theoretically for the copper-liquid helium experiments at extremely low temperatures [1–3]. However, experimental measurements of TBC were reported near room temperature at the 1980s and 1990s with the development of a transient hot-strip technique and a picosecond transient thermoreflectance (PTTR) [3,9,15,16]. As shown in Fig. 4.1 [17], after time-domain thermoreflectance (TDTR) was modified to a user-friendly version, the reported TBC values started to grow quickly. The TBC values of nonmetal-nonmetal interfaces are marked as red diamonds, while the TBC values are marked as blue squares for relatively clean metal-nonmetal interfaces including epitaxy-grown interfaces or interfaces before metal deposition cleaning steps were taken such as an in situ high-temperature bake [17]. The other TBC values of metal-nonmetal interfaces are marked as black circles. Additionally, the TBC values of two metal-metal interfaces are included in the inset of Fig. 4.1 [17].

For experimental techniques to measure TBC, a previous review paper has been summarized by Monachon et al. [12]. The 1D reference bar technique and photoacoustic technique are usually used for measuring high thermal interfacial resistances in thermal interface materials, for instance [63–65]. Electro-thermal techniques like

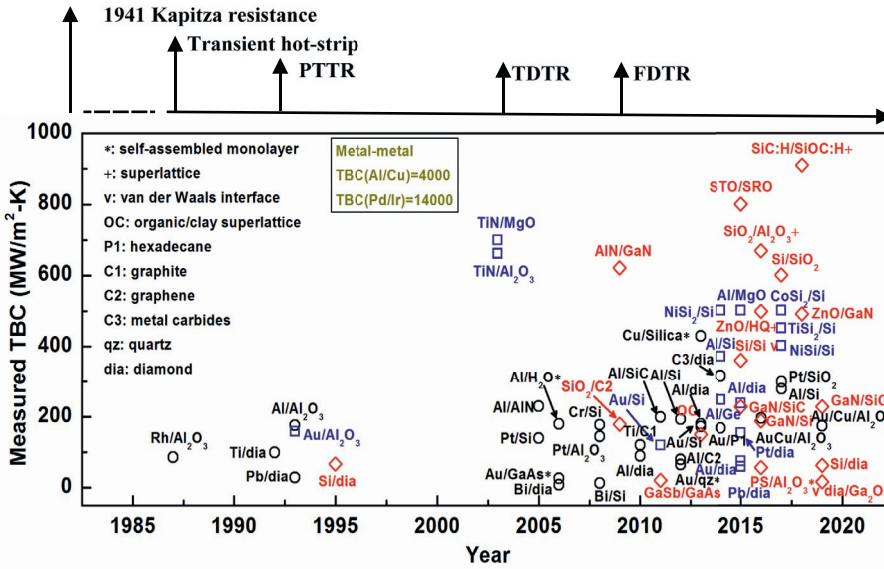


Fig. 4.1 Literature review of experimentally measured TBC at room temperature. The TBC values of nonmetal-nonmetal interfaces are marked as *red diamond*, while the TBC values of relatively clean metal-nonmetal interfaces are marked as *blue squares*, including epitaxy-grown interfaces or interfaces before metal deposition cleaning steps were taken such as an in situ high-temperature bake [17]. The TBC values of other metal-nonmetal interfaces are marked as *black circles*. Additionally, the TBC values of two metal-metal interfaces are included in the inset. TBC values (marked as *red diamond*) are Si/diamond [18], SiO_2/graphene [19], AlN/GaN [20], GaSb/GaAs [21], organic/clay superlattice [22], STO/SRO [23], GaN/SiC [24], Si/Si van der Waals interface [25], SiO_2/Al_2O_3 superlattice [26], ZnO/HQ/ZnO superlattice [27], PS/sapphire with self-assembled monolayer (SAM) [28], GaN/Si [29], SiO_2/Si [30], ZnO/GaN [6], SiC:H/SiOC:H superlattice [31], bonded GaN/SiC [14], Si/diamond [32], Ga_2O_3/diamond van der Waals interface [4]. TBC data (*blue square*) are from Au/Al_2O_3 [15], TiN/Al_2O_3 [33], TiN/MgO [33], Au/Si [34], Al/Ge [35], Al/Si [35], $NiSi_2$/Si [36], Al/MgO [23], Al/diamond [37], Pb/diamond [37], Pt/diamond [37], Au/diamond [37], NiSi/Si [38], $CoSi_2$/Si [38], $TiSi_2$/Si [38]. TBC data (*black circle*) are from Rh/Al_2O_3 [9], Ti/diamond [39], Al/Al_2O_3 [15], Pb/diamond [15], Al/AlN [40], Pt/Si [40], H_2O/Al with SAM [41], Au/GaAs with SAM [42], Bi/diamond [43], Cr/Si [44], Pt/Al_2O_3 [45], Bi/Si [46], Ti/graphite [47], Al/diamond [48], Al/SiC [49], Al/Si [50], Al/graphene [51], Au/quartz with SAM [52], Cu/silica with SAM [53], Al/O/diamond [54], Au/Ti/Si [55], metal carbides/diamond [56], Au/hexadecane [57], AuCu/sapphire [58], Al/Si [59], SiO_2/Pt [30], Au/Cu/Al_2O_3 [60]. The metal-metal TBC are from Al/Cu [61], and Pd/Ir [62].
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the 3-omega technique are usually insensitive to the TBC between the heating line and the substrates and applied to measure the thermal conductivity of substrates or thin films beneath the metal lines primarily [12,66–68]. For the 3-omega technique, the sensitivity to the metal-sample TBC is typically very low. As a result, the majority of the published TBC values of real interfaces were measured by TDTR.

After several decades' development of fundamental understanding of interfacial thermal transport, significant advances have been achieved for both experimental and theoretical studies [11,12]. However, there are several gaps in this research topic. First, most of current theoretical calculations are based on perfect interfaces that are not realistic in real-world applications, while previous experimental studies were based on imperfect interfaces. It has been reported that many factors (interfacial bonding [4,34,52], interfacial atomic mixing [44,56], pressure applied on the interfaces [37,49], surface chemistry [48,54], crystalline orientations [69,70], roughness [71], interfacial disorder [32,72]) affect TBC. Development of theoretical calculation techniques is on demand to include these factors to understand their effects on interfacial thermal transport.

Second, almost all previous experimental studies are focused on thermal measurements, while detailed interfacial structures are usually unknown or unsure whether they are as expected [73,74]. Materials characterizations, such as transmission electron microscopy, electro energy loss spectroscopy, and X-ray diffraction, are necessary to understand the atomic interfacial structure. Simultaneous thermal measurements on the same sample help relate the measured TBC with the specific interfacial structures.

Third, the nonequilibrium effect needs to be considered in the calculations of TBC. Phonons in a specific material have a large number of modes with different mean free paths. Near the interface, the temperatures of each phonon modes are different, causing strong nonequilibrium phonon transport near the interface, especially for high-mismatched heterointerfaces.

Another challenge lies on the complicated heat transport mechanisms at the interfaces that involve multiple heat carriers, as shown in Fig. 4.2 [73]. For nonmetal-nonmetal interfaces, phonons are dominant in the thermal transport via elastic and inelastic processes (Fig. 4.2A). However, for metal-nonmetal interfaces, electrons are the dominant heat carriers on one side while phonons are the dominant heat carriers on the other side (Fig. 4.2B). Thermal energy transports across the interface via multiple channels. The effects of electron-phonon interaction on the metal side and cross the interface on TBC are still on debate in the literature [12,43,75].

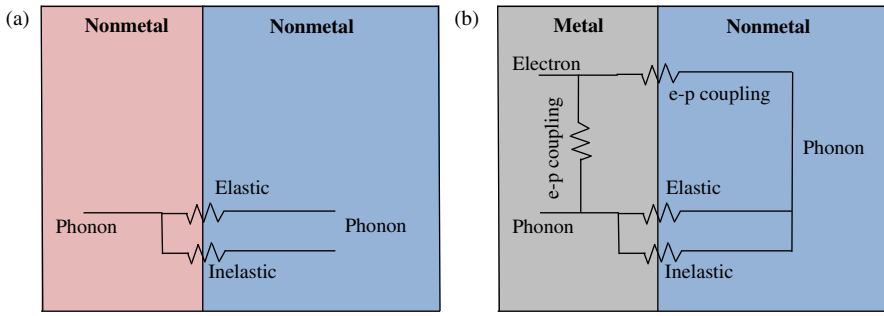


Fig. 4.2 Thermal transport mechanisms across (A) nonmetal-nonmetal interfaces and (B) metal-nonmetal interfaces.
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4.2 Thermal transport across harmonic-matched interfaces

Recently, Cheng et al. reported a systemic study of ultra-clean Al-sapphire interface with high-quality interface epitaxial growth, detailed materials characterizations, accurate thermal characterization, and newly developed modeling [73]. This work filled the gaps by providing a confirmed nearly-perfect realistic harmonic-matched interface with perfect crystal orientations which can be modeled and measured simultaneously [73].

Fig. 4.3A shows the XRD pattern of the Al-sapphire sample. The Al films are effectively single crystalline in the cross-plane direction and with twins in the in-plane direction [73]. Fig. 4.3B shows the Phi scans of the Al (220) and the sapphire (11$\underline{2}$3). The results show that the epitaxial alignment is Al [112] || Al_2O_3 [11$\underline{2}$0] which fits with ± Al [$\underline{1}$10] || Al_2O_3 [10$\underline{1}$0] [73]. The surface roughness of Al film surface,

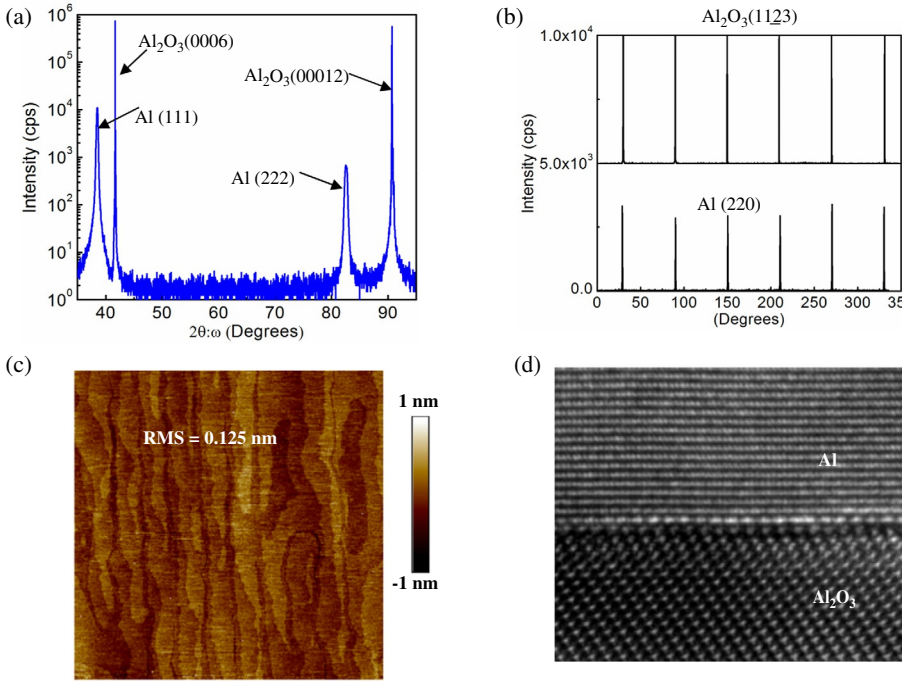


Fig. 4.3 Materials characterizations of Al-sapphire interfaces. (A) X-ray diffraction (XRD) patterns. (B) Phi scans (220) Al and (11$\underline{2}$3) Al_2O_3. (C) Atomic force microscopy (AFM) images of Al surface. The length scale bar on the image is 1 μm. Root-mean-square (RMS) roughness for a 5 μm × 5 μm area are 0.125 nm. The *color* represents surface roughness shown by the *color bar* (−1 nm to 1 nm). (D) High-resolution transmission electron microscopy (HRTEM) cross-section image of Al-sapphire interface: cross-plane [111] Al || [0001] Al_2O_3 and in-plane [110] Al || [10$\underline{1}$0] Al_2O_3. The length scale bar is 5 nm. Reproduced with permission from Z. Cheng, et al., Thermal conductance across harmonic-matched epitaxial Al-sapphire heterointerfaces, Commun. Phys. 3 (1) (2020) 1–8.

indicating atomic-level smoothness, as shown in Fig. 4.3C. The cross-section TEM image of the Al-sapphire interface shows sharp and atomic-matched interfaces, as shown in Fig. 4.3D, confirming that the high temperature annealing in ultra-high vacuum cleans the sapphire surface [73]. These detailed materials characterizations clearly confirm the high quality of the Al-sapphire interfaces. These ultra-clean interfaces provide a benchmark platform for systemically studying interfacial thermal transport by comparing experimental measurements and theoretical predictions.

As shown in Fig. 4.4A, the measured TBC of the two epitaxial grown Al-sapphire interfaces are higher than all the measured TBC values of Al-sapphire interfaces in the literature, indicating that interfacial structure affects TBC [73]. The TBC of the ultra-clean Al-sapphire interfaces have stronger temperature dependence than those of the Al-sapphire interfaces in the literature. When comparing the measured TBC with theoretically predicted TBC by nonequilibrium Landauer approach and atomic Green's function (AGF), excellent agreement is achieved, as shown in Fig. 4.4B [73]. The nonequilibrium Landauer approach corrects the temperatures of phonon modes near the interface and considers the nonequilibrium effect in the TBC calculations. The DMM is used to calculate the transmission. The AGF is also rooted on the Landauer approach but uses AGF to calculate the transmission. Both methods only include the elastic contribution to TBC. As shown in Fig. 4.2B, the Al-sapphire interfaces have three different heat transfer channels across the interface. The nearly constant TBC at high temperature shows that the inelastic contribution to TBC is not important. The excellent agreement between the measured TBC and the calculations shows that the elastic contribution to TBC is the dominant mechanism for interfacial thermal transport across Al-sapphire interfaces and the electron-phonon coupling

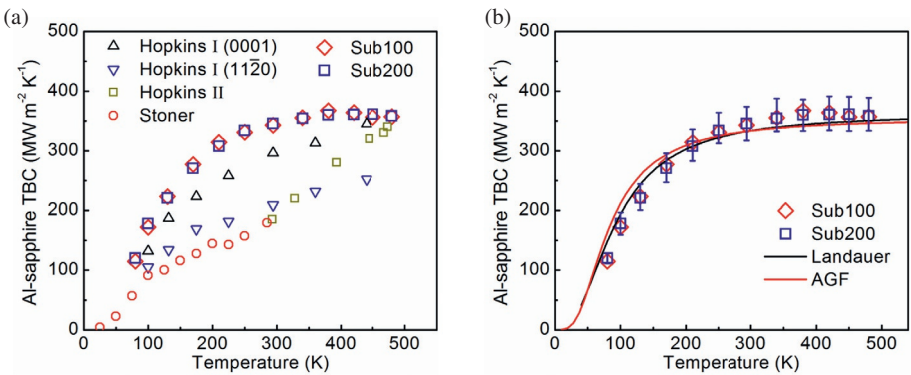


Fig. 4.4 (A) Comparison of temperature-dependent TBC of two Al-sapphire samples (Sub100 and Sub200) with measured TBC values in the literature (Hopkins I [69], Stoner [15], and Hopkins II [76]). (B) Comparison of the measured Al-sapphire TBC with the theoretical TBC values calculated by AGF and nonequilibrium Landauer approach [73].
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across the interface is negligible. This is one of the very few examples in the literature in which the measured TBC agrees well with the calculated TBC in a wide temperature range [11,38].

4.3 Inelastic contribution to TBC

At high temperatures, high-energy phonons are more likely to be excited, which facilitate the contribution of inelastic processes across interfaces. Gaskins et al. measured the TBC of epitaxial ZnO-GaN interfaces over a wide range of temperatures [6]. Fig. 4.5A and B shows the AFM-measured surface images of the ZnO and GaN. The surface is smooth with a roughness value of a few nm. Fig. 4.5C shows the cross-section TEM image of the ZnO-GaN interfaces. A dashed yellow line is used to mark the interface. Unlike the Al-sapphire interfaces, the ZnO-GaN interfaces are not atomic sharp and there are large amount of defects in the ZnO layer near the interface [6].

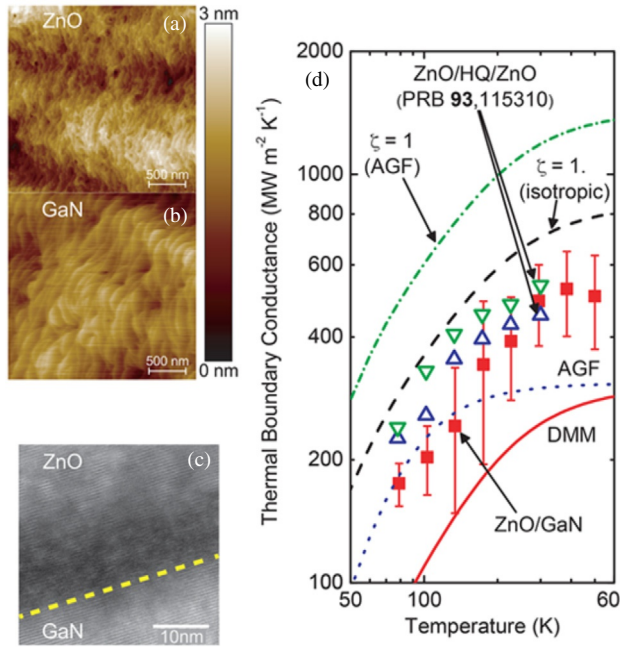


Fig. 4.5 (A) AFM image for the ZnO. (B) AFM image for the GaN wafer. (C) Cross-section HRTEM image of the ZnO-GaN interfaces. The dashed yellow line shows the interface. (D) The measured TBC of the epitaxial grown ZnO-GaN interfaces. DMM-based Landauer approach and AGF calculated TBC are included as comparison. Here, the DMM-based Landauer approach does not consider the nonequilibrium effect.
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The original version of the Landauer approach with DMM and AGF is used to calculate the ZnO-GaN TBC, which are included in Fig. 4.5D for comparison [6]. The DMM-based Landauer approach does not include the nonequilibrium effect. As a result, the measured TBC is much higher than the TBC calculated by the Landauer approach, even at low temperatures at which elastic contribution should dominate interfacial thermal transport. The AGF-calculated TBC matches well with the measured TBC at low temperatures, indicating the elastic processes dominate the measured TBC at low temperatures. However, at high temperatures (above room temperature), the measured TBC are higher than the AGF-calculated TBC, indicating inelastic contributions to TBC start to make a difference. For the ZnO-GaN interfaces, these harmonic based approaches cannot capture the effect of anharmonicity to TBC at high temperatures. Additionally, by assuming the phonon transmission coefficient as unity, the TBC based on the Landauer formula and AGF, which are the upper limit of the elastic contributions, are higher than the measured TBC.

Similar observations have been reported for GaN-AlN interfaces as well [77]. The TBC of GaN-AlN interfaces calculated by a nonequilibrium Green's function formalism in the harmonic limit is only half of the measured TBC interred from GaN-AlN superlattices at room temperature, indicating large contributions from the inelastic processes. Current state of art computational approaches for TBC calculations are usually limited to include inelastic contributions in the TBC calculations [78]. Molecular dynamics simulations can capture the inelastic contributions well but are usually limited by accurate atomic potentials, especially for materials with complex crystal structures.

4.4 Effects of interfacial bonding to TBC

Interfacial bonding tuned by interfacial chemistry, types of bonding, and applied pressure has been reported to impact interfacial thermal transport [34,49,52]. By tuning the interfacial chemistry via self-assembled monolayer, the TBC was found to be related to chemical bonding [52]. The external pressure applied on the interface was also studied to tune the TBC of weak interfaces [49]. Recently, the TBC of GaN-diamond interfaces is reported to be tuned for orders of magnitude by varying the adhesion energy theoretically [79]. For a direct van der Waals bonded GaN-diamond interface, the measured TBC is as low as $5\,MW/m^2\,K$, which is 18 times lower than the TBC of a bonded GaN-diamond interface with a 4-nm-thick interfacial layer [79,80]. The TBC of van der Waals bonded β-Ga_2O_3-diamond interface is around $17\,MW/m^2\,K$ at room temperature, which is one order of magnitude lower than that of the atomic layer deposited β-Ga_2O_3-diamond interface [4,81].

4.5 Comparison of TBC modeling methods

To model and simulate the thermal transport across interfaces, there are two main categories of methods. One is usually called the phonon gas model and based on the Landauer formula with phonon interfacial transmission functions calculated from

different approaches, as we discussed above. The other is within the framework of molecular dynamics (MD) [82–84]. More recently, interface conductance modal analysis (ICMA) [85,86] based on MD was developed to predict TBC across interfaces.

The idea of the phonon gas model is that considering the interfacial energy transport process as the phonon transmission process, and the net heat flux at the interface is the total incident energy of phonons from material 1 times the transmission coefficient at the interface from material 1 to 2 subtracted by the transmitted heat flux from 2 to 1. From first-principle calculations or empirical interatomic potentials, the total incident energy of phonons can be obtained from the integral the energy of each phonon over all incident phonon modes. The calculations of the transmission coefficients are where the difference of TBC values come from for thermal transport from material 1 to different materials.

To capture the transmission across interfaces, AMM, DMM, AGF, and phonon wave-packet method [3,5,87–91] are common methods to predict the transmission coefficient. It should be noted that the phonon wave-packet method usually needs to be applied within the framework of MD to simulate the transmission of a modal wave-packet, while the transmission of other three methods can be obtained directly from first-principle calculations or empirical interatomic potentials information.

Among these four methods, the advantages of AMM and DMM are that the computational cost is low for relatively good agreement between modeling and experiments in a limited number of cases [13,73]. AMM usually works well for smooth interface since it is assumed with complete specularity that the phonons are governed by continuum acoustics and the interface is treated as a plane [3], while DMM assumes the opposite that all the phonons are diffusely scattered at the interface. However, for complex interfaces, these two methods cannot capture all features of the phonon transport across the interface. The values predicted from AMM and DMM can be considered as the limits of the TBC. Compared with AMM and DMM methods, the advantage of AGF and phonon wave-packet methods is that they can easily capture the detailed atomic structures or different interface bonding strengths of the interface. For the AGF method, the calculation of transmission does not rely on interatomic potentials and can be obtained from first-principle calculations. For the phonon wave-packet method, it is easy to understand the contribution of different phonon modes to the interfacial thermal transport since the wave-packet of a single phonon mode is launched in each simulation.

Within the framework of MD methods, nonequilibrium MD [82–84] and ICMA [85,86] are usually applied to predict TBC. The anharmonicity of lattice vibrations can be easily included from the higher order force constants of empirical interatomic potentials within the framework of MD, while it is usually hard to include the effect of anharmonicity in the phonon gas model. There are some recent efforts to include the anharmonicity in the AGF interfacial phonon transmission calculations [78,89]. However, there are still limitations, such as inaccurate estimation of the interfacial phonon scattering rates, limited simulated structures, or the extremely high computational costs.

Moreover, the interface structures in MD simulations are quite flexible that complex interfacial details (e.g., strong interfacial disorder and interfaces with

dimensional mismatch) can be simulated by creating different initial atomic structures. Also, the local nonequilibrium at the interface can be easily captured within the framework of MD because of the way temperature is defined in MD. However, the nonequilibrium effect is originally not considered in Landauer-based calculations, which leads to finite thermal conductance across imaginary interfaces. Recently, there are some progress considering the interfacial nonequilibrium with phonon gas model and a better agreement with experiments have been achieved with the interfacial temperature correction [8].

However, MD is a classical method, and quantum effects are not considered in MD simulations. Although there are some methods to do the quantum corrections [92], there are still limitations and inaccuracies especially for the simulations at low temperature or with confinements. For the phonon gas model, the quantum effect can be included from the correct carrier statistics, which is significant at sub-Debye temperatures especially for high Debye temperature materials such as diamond and GaN. Another limitation of the MD-based methods is that the interatomic potentials are necessary but difficult to obtain. The MD-based methods cannot be applied to systems without appropriate potentials. While for the phonon gas model with transmission from AMM, DMM, and AGF, the transmission functions can be obtained from first-principle results.

Acknowledgments

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